Akshat Sachin Valse

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Stochastic analysis for machine learning: Langevin sampling and its discretization, large deviations and rare-event simulation, the stability of learned PDE solvers, and the queueing, calibration and reliability questions raised by systems that run LLMs.

Education

Iowa State University, Honors Program Expected May 2028
B.S. Statistics · B.S. Mathematics · B.A. Computer Science · B.S. Data Science
Cumulative GPA 3.86 / 4.00.

In progress, Fall 2026 (all four at the PhD level, 6000-series). Foundations of Probability Theory, measure-theoretic (STAT 6541); Advanced Topics in Machine Learning (COM S 6730); Advanced Topics in Artificial Intelligence (COM S 6720); Advanced Topics in High-Performance Computing (COM S 6250).

Planned, Spring 2027. Advanced Probability Theory; Advanced Stochastic Processes.

Graduate and advanced coursework, completed.
Probability and statistics: Theory of Probability and Statistics (STAT 5430); Stochastic Process Models (STAT 5540); Empirical Methods for the Computational Sciences (STAT 5830); Multivariate Analysis (STAT 5750); Categorical Data Analysis (STAT 5770); Statistical Computing (STAT 5860).
Mathematics: Numerical Methods for Differential Equations (MATH 5810); Analysis I (MATH 4140); Introduction to Mathematical Logic (MATH 4430X); Partial Differential Equations (MATH 3850); Theory of Linear Algebra, honors (MATH 3170).
Computing and machine learning: Concepts and Applications of Machine Learning, honors (DS 3030); Design and Analysis of Algorithms (COM S 3110); Software Development Practices (COM S 3090).

Research Experience

Langevin Sampling and Diffusion Estimation for Molecular Systems May 2026 – present
Research Collaborator, Department of Mathematics, Iowa State University. Advisor: Prof. David Herzog. Funded by NSF award DMS-2246491.

- Built an end-to-end Python simulation pipeline implementing MALA and projected Euler–Maruyama samplers for overdamped Langevin dynamics, staged from a one-dimensional periodic potential to a solvated ion in a three-dimensional Lennard-Jones bath; estimated self-diffusion and mobility coefficients from long-run trajectory data.
- Derived the closed-form Lifson–Jackson diffusivity $D = 1/I_0(1)^2 \approx 0.62386$ as an exact benchmark and validated the estimators against it before scaling to three dimensions, where the pipeline reproduces published solvated-ion results.
- Author weekly technical reports, including a companion note supporting the group's CLT-based error analysis.

Large Deviation Theory for Rare-Event Analysis in Queueing Systems Nov. 2024 – present
Undergraduate Research Fellow, Department of Mathematics, Iowa State University. Advisor: Prof. Ruoyu Wu. Dean's High Impact Undergraduate Research Award.

- Developed the first importance-sampling methodology for rare-event estimation in the mean-field join-the-shortest-queue model, deriving state-dependent exponential tilting from the Budhiraja–Friedlander–Wu (2021) large-deviation rate function; stated a logarithmic-efficiency conjecture with a proof sketch via the Dupuis–Wang subsolution framework.

- Built a reproducible, performance-tuned Python simulation framework (Xoshiro256** with deterministic seeding) and validated it across 27 configurations ($n \leq 100$, $\rho \in \{0.7, 0.9, 0.95\}$): estimates agree with crude Monte Carlo within 95% confidence intervals wherever crude Monte Carlo succeeds, and extend to probabilities as small as 10^{-31}, where it fails entirely.
- Designed and benchmarked two heuristic tilting families (rate-reversal, level-weighted) with pilot-run selection; diagnosed over-tilting at extreme rarity through likelihood-ratio calibration statistics, motivating the ongoing computation of the optimal state-dependent tilt.

Stability Analysis of Cell-Average Neural Networks for PDEs May 2026 – present

Undergraduate Researcher, Department of Mathematics, Iowa State University. Advisor: Prof. Jue Yan. Funded by the NSF D4 NRT. Joint with Mengyuan Yang.

- Adapting von Neumann stability analysis to neural-network PDE solvers: deriving the linearized amplification factor of the learned time-stepping scheme and characterizing how perturbations propagate through it, toward stability conditions on the trained weights analogous to those of classical finite-difference methods.
- Reimplementing the group's solver in Python to enable systematic stability experiments across schemes and step sizes.

Metastable Failures in Prefix-Cached LLM Agent Serving Aug. 2026 – present

Research project, Advanced Topics in High-Performance Computing (COM S 6250), Department of Computer Science, Iowa State University. Advisor: Prof. Ali Jannesari, SwaPP Lab.

- Asking whether a short burst of agent traffic can push a healthy LLM serving engine into a self-sustaining all-miss, low-throughput state, and which scheduling and admission mechanisms prevent it. Built a discrete-event simulator of a continuous-batching engine with prefix caching (scheduling policies, eviction rules, memory-allocation modes, and re-implementations of recently proposed mechanisms) and a queueing-theoretic "thrashing curve" model of when caching helps, when it collapses, and when admission control restores stability.
- Run real-engine experiments (SGLang serving a 7B-parameter model on NVIDIA A100 GPUs) on Iowa State's Nova HPC cluster through SLURM, with predictions registered before each run, paired control arms, and every reported number traced to a results directory, a commit and a cluster job ID. Toward an arXiv preprint.

Calibrating Volatility Forecasts with Early, One-Sided Feedback Aug. 2026 – present

Research project, Advanced Topics in Machine Learning (COM S 6730), Department of Computer Science, Iowa State University. Advisor: Prof. Hongyang Gao.

- A forecast of a cumulative quantity is scored only when its horizon ends, but its increments are non-negative, so a miss becomes certain the moment the running sum crosses the issued bound. Designed a model-agnostic calibration layer over any quantile forecaster that credits those certain early misses and keeps explicit accounts of forecasts still pending; proved its long-run coverage guarantee under early crediting, bounded correction and missing data, and checked it numerically over tens of thousands of adversarial runs.
- Built a leakage-audited data pipeline (5-minute spot bars, 2020–2025, with a sealed and hashed test set), a regime-shift simulation study with exact oracle quantiles and pre-registered hypotheses, and delayed-feedback conformal and classical volatility baselines, including a temporal convolutional network trained on HPC GPUs. Paper in preparation.

Reliability of Tool-Using LLM Agents Aug. 2026 – present

Research project, Advanced Topics in Artificial Intelligence (COM S 6720), Department of Computer Science, Iowa State University. Advisor: Prof. Yang Li.

- When an agent stalls, when is restarting it worth more than waiting, and how can the probability of a failure too rare for direct simulation be estimated? Two pre-registered pilots on real agents, one on restart policies and one on rare failures, with the direction still being chosen.

Manuscripts in Preparation

1. *Large-Deviation-Guided Importance Sampling for Rare Events in the Mean-Field Join-the-Shortest-Queue Model.* With R. Wu.
2. *Langevin Dynamics and Transport-Coefficient Estimation for Molecular Systems.* With D. Herzog.
3. *Stability Certificates for Cell-Average-Based Neural Network Solvers: von Neumann Analysis of Trained Flux Networks.* With M. Yang and J. Yan.
4. *Metastable Failures in Prefix-Cached LLM Agent Serving.* With A. Jannesari.
5. *Online Calibration of Neural Volatility Forecasts with Early, One-Sided Feedback.* With H. Gao.

Awards and Honors

- Dean's High Impact Undergraduate Research Award, Iowa State University, 2026.
- Schillmoeller Family Scholarship in Statistics.
- Mead Fund Award in Statistics.
- James L. Cornette Mathematics and Interdisciplinary Studies Award.
- Dean's List, every semester.

Leadership and Service

- Treasurer, Iowa State Data Science Club, 2025 – present. Secured $60,000+ in funding supporting 130+ members.
- Research Ambassador, Iowa State University Honors Program.
- FAA Private Pilot Certificate, high-performance endorsement, 2025. Volunteer pilot and recruitment advisor, Pilots N Paws animal-rescue flights.

Technical Skills

Languages and tools. PyTorch and CUDA for neural-network and GPU-accelerated simulation; Python (NumPy, SciPy, pandas, scikit-learn, matplotlib, asyncio); SLURM on GPU clusters and LLM serving engines (SGLang); R; LaTeX; git.

Methods. Measure-theoretic probability and stochastic differential equations; MCMC (MALA, Metropolis–Hastings) and importance sampling; large-deviation asymptotics; numerical analysis of time-stepping schemes and von Neumann stability; discrete-event simulation and queueing models; online conformal calibration; reproducible computation with recorded seeds, parameter sidecars and pre-registered predictions, so every figure regenerates from saved results.